INCH-POUND
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MILITARY SPECIFICATION

COPPER ALLOY INVESTMENT CASTINGS

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers copper base alloy castings made by the investment casting process.

1.2 Classification. Copper base alloy investment castings shall be of the compositions and conditions shown in table I, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENES, Wright-Patterson AFB OH 45433-6503 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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TABLE I. Compositions and Conditions

Copper Alloy no.	Former composition no.	Description	Condition
825	10	Beryllium copper	A. As cast B. As cast & precipitation hardened C. Solution annealed D. Solution annealed & precipitation hardened
836	2	Ounce metal	As cast
861	7	Aluminum manganese bronze	As cast
863	9	Aluminum manganese bronze	As cast
865	5	Manganese bronze	As cast
874	4	Silicon brass	As cast
903	3	Phosphor bronze	As cast
934	1	Bearing bronze	As cast
952	6	Aluminum bronze	A. As cast
955	8	Aluminum nickel bronze	B. Heat treated

SPECIFICATIONS

FEDERAL

UU-P-268	Paper, Kraft, Untreated, Wrapping
MMM-A-260	Adhesive, Water-Resistant, (For Sealing Waterproofed Paper)
PPP-B-566	Box, Folding, Paperboard
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber
PPP-B-665	Box, Paperboard, Metal-Stayed, (including Stay Material)

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PPP-B-676	Box, Setup
PPP-B-1055	Barrier Material, Waterproofed, Flexible
PPP-C-1120	Cushioning Material, Uncompressed Bound Fiber for Packing
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, for Carbon Sealing

MILITARY

MIL-P-116	Preservation, Methods of

STANDARDS

FEDERAL

FED-STD-151	Metals, Test Methods

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-271	Nondestructive Testing Requirements for Metals
MIL-STD-278	Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping and Press Vessels in Ships of U.S. Navy

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E8	Tension Testing of Metallic Materials
ASTM E18	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Chicago Union Station, Chicago IL 60606.)

AMERICAN NATIONAL STANDARDS

ANSI B46.1 Surface Texture

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Process. Castings shall be made by any investment process producing castings complying with this specification.

3.2 Chemical composition. The chemical analyses of castings shall comply with table II, for the composition as specified (see 6.2).

3.2.1 The contractor shall furnish an analysis of each master heat showing the percentages of the elements designated. A master heat shall be defined as previously refined metal of a single furnace charge or a blend thereof.

3.3 Condition. Castings shall be furnished in the condition specified in the contract or order (see 6.2). The type of heat treatment, when required, shall be at the option of the contractor unless otherwise specified by the procuring activity.

3.4 Mechanical properties. Mechanical properties shall be as shown in table III. Mechanical properties of castings required for conditions other than shown in table III shall be as specified in the contract or order (see 6.2).

3.5 Soundness

3.5.1 Surface. Castings shall be free from cold shuts, fins, laps, or other defects detrimental to fabrication or performance of parts.

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TABLE II. Chemical Requirements[1]

(Composition, percent maximum except as indicated)

Copper alloy no.	Former composition no.	Copper[2]	Tin	Lead	Zinc[2]	Iron	Antimony	Nickel	Sulfur	Phosphorous	Aluminum	Manganese	Silicon	Beryllium	Cobalt	Chromium
825	10	Remainder	.10	.02	.10	.25	-	.20[3]	-	-	.15	-	.20-.35	1.90-2.15	.35-.65[4]	.10
836	2	84.0-86.0[5]	4.0-6.0	4.0-6.0	4.0-6.0	.30	-	1.0[3]	-	0.5[6]	-	-	-	-	-	-
861	7	66.0-68.0	.20	.20	Remainder	2.0-4.0	-	-	-	-	4.5-5.5	2.5-5.0	-	-	-	-
863	9	60.0-66.0	.20	.20	Remainder	2.0-4.0	-	-	-	-	5.0-7.5	2.5-5.0	-	-	-	-
865	5	55.0-60.0	1.0	.40	Remainder	.40-2.0	-	.50	-	-	.50-1.5	1.5-	-	-	-	-
874	4	99.5 min[7]	-	1.0	12.0-16.0	-	-	-	-		-		2.5-4.0	-	-	-
903	3	86.0-89.0	7.5-9.0	.30	3.0-5.0	.15	-	1.0	-	0.5[6]	-	-	-	-	-	-
934	1	82.0-85.0	7.0-9.0	7.0-9.0	.7	.15	.50	1.0	-	.50[6]	-	-	-	-	-	-
952	6	86.0 min[8]	-	-	-	2.5-4.0	-	-		-	8.5-9.5	-	-	-	-	-
955	8	78.0 min[8]	-	-	-	3.0-5.0	-	3.0-5.5	-	-	10.0-11.5	3.5	-	-	-	-

[1] By agreement between consumer and producer, analyses may be required and limits established for chemicals not specified in the table of chemical requirements.

[2] In reporting chemical analyses obtained by the use of instruments, such as spectrograph, X-ray, and atomic absorption, copper may be indicated as "remainder".

In reporting chemical analyses obtained by wet methods, zinc may be indicated as " remainder" on those alloys with over 2%.

[3] Nickel is a residual element and shall not be intentionally added to the melt.

[4] Nickel plus cobalt

[5] In determining copper minimum, copper may be calculated as copper plus nickel.

[6] For continuous castings, the maximum phosphorus shall be 1.9%

[7] Copper plus sum of named elements

[8] "Total named elements " 99.0% min.

[9] "Total named elements" 99.9% min.

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TABLE III. Mechanical Properties

(As cast condition except where noted)

Copper alloy no.	Former composition no.	Tensile strength (ksi-min)	Yield strength[1] (ksi-min)	Elongation (in 2 inches - min)	Hardness (Rockwell or equivalent)
825	10	70.0	-	15	B75 min.
825 (heat treated and aged)	10	155.0	140.0	0	C36 min.
825 (as cast and aged)	10	85.0	-	15	B90 min.
825 (as cast solution heat treated[2] and water quenched)	10	55.0 max.	-	30 max.	B60 - 75
825 (as cast solution heat treated[2] and water quenched and precipitation[3] hardened)	10	155.0	-	0	C38 min.
836	2	30.0	14.0	20	-
861	7	90.0	45.0	18	-
863	9	110.0	60.0	12	-
865	5	65.0	25.0	20	-
874	4	50.0	21.0	18	-
903	3	40.0	18.0	20	-
934	1	25.0	12.0	8	-
952	6	65.0	25.0	20	-
955	8	90.0	40.0	6	-
955 (heat treated)	8	110.0	60.0	5	-

[1] Yield strength taken at 0.5% extension under load except for copper alloy No. 825 (heat treated) which is to be taken at 0.2% effort.

[2] Solution heat treatment consist of heating the castings for at least 1 hour at 1475° ±10°F followed by an immediate water quench.

[3] Precipitation hardening consist of heating the castings for at least 2 hours at 650° ±10°F, followed by air cooling. Other temperature-time cycles can be used to achieve the minimum properties of this specification.

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3.5.2 Internal. Castings shall conform to MIL–STD–278 with criticality levels for bronze castings as specified. Where level is not specified, level 3 shall apply.

3.6 Repairing of defects. Castings shall not be welded, plugged, or repaired in any manner without written permission from the procuring activity. Permission to repair defects, when granted, shall not relieve the contractor from his responsibility to meet all the requirements of the drawings and the specification.

3.7 Dimensions and surface roughness. Castings shall conform to the drawings specified in the contact or order (see 6.2) with respect to dimensions and tolerances. Unless otherwise specified, roughness of surfaces shall not exceed 125 rha. Surface roughness shall be interpreted in accordance with the provisions of ANSI B46.1.

3.8 Identification marking. Castings shall be identified with the melt or inspection lot number, and in addition, when specified on the drawing, shall be marked with the drawing or part number by any method which has no deleterious effect on serviceability.

3.9 Workmanship. Castings shall be free from foreign materials and other injurious defects which may adversely affect its intended use. Castings shall be well cleaned. Unless otherwise specified, metallic shot or grit shall not be used for final cleaning.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of *sections* 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Sampling

4.2.1 Lot.

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4.2.1.1 Lot for chemical analysis. For purposes of chemical analysis, a lot shall consist of all castings as poured from the same master heat submitted for inspection at one time.

4.2.1.2 Lot for soundness, hardness, tensile, surface roughness, workmanship and dimensional inspections. A lot shall consist of all castings of the same drawing, from the same master heat, and in the same condition submitted for inspection at one time.

4.2.1.3 Castings shall be properly separated by lots when presented for Government acceptance.

4.2.2 Sampling for lot acceptance inspection

4.2.2.1 Sampling for visual and dimensional examination. Sampling of castings for dimensional, identification marking, and workmanship inspection shall be in accordance with the provisions of MIL-STD-105, inspection level II, and acceptable quality level (AQL) 1.5 percent defective..

4.2.2.2 Sampling for soundness. Sampling of castings for soundness inspection shall be in accordance with the provisions of MIL-STD-105, inspection level II, and AQL 0.65 percent defective.

4.2.2.3 Sampling for packaging, packing, and marking. A random sample of shipping containers for the inspection of 4.3.3 shall be selected in accordance with the provisions of MIL-STD-105, inspection level II, AQL 4.0 percent defective.

4.2.2.4 Sampling for tests.

4.2.2.4.1 For chemical analysis. A sample for chemical analysis shall be taken from each lot. The sample may be taken from test coupons or from representative castings.

4.2.2.4.1.1 Wet chemical analysis. The sample shall consist of not less than 2 ounces of drillings taken from sound metal below the surface from the cross-section in the cast and shall be fine and clean, free from oil, grit, and other foreign material.

4.2.2.4.1.1.1 Spectrochemical analysis. Samples shall be obtained as specified in method 112 of FED-STD-151.

4.2.2.4.1.1.1.1 Other methods of analysis. Samples for other methods of analysis shall be as agreed upon by the contractor and the procuring agency.

4.2.2.4.2 For tension tests. Three test bars shall be poured to represent each lot of castings. Test bars shall be separately cast from the same master heat and under the same conditions as the lot of castings to be represented. They shall be made in molds of the same material and cooled at approximately the the same rate as the molds for the lot of castings to be represented. Each test bar shall have a gage section diameter of 0.250 ± 0.010 inch and a gage length of not less than 1 inch. The radius of the fillet shall be not less than 1/4 inch. Each test bar shall be marked for future identification.

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4.2.2.4.3 *For surface roughness and hardness.* Sampling of castings for surface roughness and hardness shall be in accordance with the provisions of MIL–STD–105, inspection level II, and AQL 1.5 percent defective.

4.3 Lot acceptance inspection.

4.3.1 Visual and dimensional examination. Sample castings selected in accordance with 4.2.2.1 shall be inspected to determine conformance to the dimensional requirements of 3.7, the identification marking of 3.8, and the workmanship requirements of 3.9.

4.3.2 Soundness inspection. Sample castings selected in accordance with 4.2.2.2 shall be subjected to penetrant inspection and to radiographic inspection in accordance with MIL–STD–271 to determine conformance to the soundness requirements of 3.5.

4.3.3 Preservation, packaging, packing, and marking. Sample shipping containers shall be inspected to determine that preservation, packaging, packing, and marking of containers selected in accordance with 4.2.2.3 are in accordance with the requirements of this specification and the contract or order.

4.3.4 Tests. The methods of test for chemical composition, tension tests, and hardness tests shall be in accordance with FED–STD–151.

4.3.4.1 Chemical composition. Unless otherwise specified the metal sample selected in accordance with 4.2.2.4.1 shall be analyzed by method 111 of FED–STD–151 to determine conformance to the requirements of 3.2. If any sample fails to conform to these chemical requirements, the lot represented by the sample shall be rejected.

4.3.4.2 *Tension tests.* Test bars prepared in accordance with 4.2.2.4.2 shall be pulled in tension in accordance with the method specified in 4.3.4.2.1 and ASTM E 8 to determine conformance to the mechanical properties of 3.4. If a test bar fails to conform to these mechanical properties, the lot represented by the test bar shall be subject to rejection. Retests will be permitted in accordance with the provisions of FED–STD–151.

4.3.4.2.1 Method of tension test. Separately cast bars shall be tested without machining except as may be necessary to remove gates and to adapt the ends to the grips of the testing machine. Strength shall be computed from the actual measured diameter of the test bar before testing.

4.3.4.3 Surface roughness. Sample castings selected in accordance with 4.2.2.4.3 shall be inspected to determine conformance to 3.7.

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4.3.4.4 Hardness test. Each casting selected in accordance with 4.2.2.4.3 shall be subjected to a hardness test in accordance with ASTM E 18 to determine conformance to the requirements of 3.4. Any casting that fails to meet the hardness requirements shall be rejected and if the number of castings rejected exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

4.4 Disposition of nonconforming product. Rejected lots may be offered again for inspection provided the contractor has removed all nonconforming castings in accordance with MIL-STD-105. The inspector shall again select samples from resubmitted lots, as provided in 4.2, to verify compliance with the applicable specification requirements.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2)..

5.1.1 Level A.

5.1.1.1 Cleaning. Unless otherwise specified, castings shall be cleaned in accordance with method C-1 of MIL-P-116.

5.1.1.2 Drying. Unless otherwise specified, castings shall be dried in accordance with one or more of the applicable procedures of MIL-P-116 provided the procedure is not injurious to the item.

51.1.3 Unit packaging. Castings shall be unit packaged in accordance with method III of MIL-P-116. The unit package shall contain one casting wrapped in two thicknesses of 40-pound kraft paper conforming to grade B of UU-P-268.

5.1.1.4 Intermediate packaging. When specified in the contract or order (see 6.2), unit packages shall be further packaged in folding boxes, set-up boxes, metal-stayed boxes, or fiber boxes conforming to PPP-B-566, PPP-B-676, PPP-B-665, or PPP-B-636. All seams and joints, including manufacturer's joints, shall be covered with water-resistant tape conforming to PPP-T-76. Gross weight of packages shall not exceed 10 pounds for PPP-B-566 and PPP-B-676 or 20 pounds for PPP-B-636 and PPP-B-665.

5.1.1.5 Cushioning. Castings shall be protected on all sides with sufficient thickness of cushioning material conforming to PPP-C-1120. At the time of packaging and packing, the moisture content of the cushioning materials shall be no more than 5 percent. The pH range of the material shall be between 6.0 and 8.0.

5.1.2 Level C. Castings shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

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5.2.1 Packages per pack. The number of packages per pack shall be as specified in contract or order (See 6.2).

5.2.2 Level A. Castings, packaged as described in 5.1.1, shall be packed in cleated plywood boxes, nailed wood boxes, or wirebound boxes conforming to PPP-B-601 (overseas type), PPP-B-621 (class 2) and PPP-B-585 (class 3) , respectively. Shipped containers shall be lined with a sealed waterproof bag, or its equivalent, made of material conforming to PPP-B-1055, for case liners. The seams and closures shall be sealed with water-resistant material conforming to the requirements of MMM-A-260. The gross shipping weight shall not exceed 150 pounds.

5.2.3 Level B. Castings, packaged as described in 5.1.1, shall be packed in wood boxes conforming PPP-B-585 (class 1), PPP-B-601 (domestic type), PPP-B-621 (class 1). The gross shipping weight shall not exceed 150 pounds for wood boxes and 70 pounds for fiber boxes.

5.2.4 Level C. Castings, packaged in accordance with 5.1.2, shall be packed or otherwise prepared for shipment to insure carrier acceptance and to insure safe delivery to destination at the lowest applicable rate. Shipping containers shall conform to the Uniform Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2), packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Castings purchased under this specification are intended primarily for use in manufacture and assembly of ordnance equipment which requires small, sound, high quality, and intricate precision castings weighing from approximately 1 ounce to 6 pound.

6.1.1 Copper alloy No. 825 and 874 (composition 10 and 4) have better castability than other copper-base alloys. They can be cast more readily into intricate shapes with better surface finishes. Copper alloy No. 934, 836, and 903 (compositions 1,2, and 3) are more difficult to cast than copper alloy No. 825 and 874, (composition 10 and 4) but are superior to copper alloy No. 865, 952, 861, 955, and 863 (compositions 5 through 9).

6.1.2 Copper alloy No. 934 (composition 1) possess fair strength and is suitable for parts in which bearing surfaces are required or pressure tightness is required.

6.1.3 Copper alloy No. 836 (composition 2) is used in castings requiring fair strength and soundness. It is suitable for fittings in hydraulic systems and small gears.

6.1.4 Copper alloy No. 903 (composition 3) may be used for special pipe fittings expansion joints, gears, valves, pump components, bushings, and similar applications requiring good strength and resistance to sea water corrosion.

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6.1.5 Copper alloy No. 874 (composition 4) in addition to being an excellent casting alloy has high strength and corrosion resistance. This alloy is not recommended for high pressure steam service.

6.1.6 Copper alloy No. 865 (composition 5) should be used for gears, worm wheels, framing, etc., where strength, toughness, and resistance to sea water corrosion is required. Superior resistance to corrosion by dezincification can be achieved by specifying a minimum of 0.50 percent tin in the contract or order when purchasing castings of this composition.

6.1.7 Copper alloy No. 952 and 955 (composition 6 and 8) are aluminum bronzes and the most difficult alloys to cast because of their dross forming tendency. However, their superior corrosion resistance and excellent strength properties can be used to advantage at some sacrifice in surface finish and intricacy of design. These alloys are used for worm gears, valve seats, bearings, and other applications where corrosion, shock, and fatigue are encountered.

6.1.8 Copper alloy No. 861 and 863 (composition 7 and 9) are intended for castings such as gears, worm wheels, framing, etc., where strength, toughness, or resistance to sea water corrosion is required. When stress corrosion is a factor, copper alloy No. 861 (composition 7) should used.

6.1.9 Copper alloy No. 825 (composition 10) is one of the more widely used copper alloys for investment castings because of its high mechanical properties. It should be noted, however, that the properties of copper alloy No. 825, condition B, can be obtained only by proper heat treatment which requires quenching from solution treating temperatures and aging. Hence, distortion may be encountered during heat treatment obtaining the maximum mechanical properties.

6.2 Acquisition requirements. Acquisition requirements documents should specify the following:

a. Title, number, and date of this specification.

b. The composition and condition of castings required (see 1.2, 3.2, and 3.3).

c. Mechanical properties when required (see 3.4).

d. Applicable drawings or dimensions of castings (see 3.7).

e. Place of inspection and test, if different from 4.3.4 and 4.3.4.1.

f. Level of packaging (see 5.1).

g. Intermediate packaging, when required (see 5.1.14).

h. Level of packing (See 5.2).

i. Number of packages per pack required (see 5.2.1).

j. Additional marking, if required (see 5.3).

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6.3 Preparation for delivery criteria. Criteria for use of the proper level of packaging and packing shall be as follows:

Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

Level B. This level shall be used for protection against damage during multiple domestic shipments, handling, and covered storage

Level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

6.4 Subject term (key word) listing

Castings
Composition
Copper alloy

6.5 Change from previous issue. *Marginal notations* not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force – 11

Review activities:
Air Force – 99

Preparing activity:
Air Force – 11

Proj – MECA-F030

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, not to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractural requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-22087B(USAF)	2. DOCUMENT DATE *(YYMMDD)* 930615

3. DOCUMENT TITLE
Copper Alloy Investment Castings

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

A. NAME	B. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*
C. ADDRESS *(Include Zip Code)*	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: *Defense Quality and Standardization Office* 5203 Leesburg Pike, Suite 1403, Falls Church VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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